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Mass and efficiency optimization of a He-Xe Brayton cycle for mobile, land-based nuclear microreactors

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ABSTRACT

Nuclear microreactors represent an option for distributed power generation, which can be deployed in military installations, remote off-grid villages, and locations recovering from natural disasters. Small system mass is necessary to achieve their mobility. In this study, a He-Xe gas-cooled reactor coupled with a Closed Brayton cycle (CBC) is optimized in terms of system mass and thermal efficiency. A single-channel model is developed for the reactor to predict the minimum mass according to the thermal–hydraulic limitations. Masses of the heat exchanger and turbo-machines are calculated based on empirical curves. The simple regenerative and inter-cooling cycles are considered. A parametric study is conducted to investigate the influence of key parameters on system mass and to explore the tradeoffs between different components that satisfy the power requirements of 8 MWe. Furthermore, the Non-dominated Sorting Genetic Algorithm (NSGA-II) is used to optimize the system mass and thermal efficiency. Optimal values of critical parameters on the Pareto Frontier are presented and discussed. Results show that the masses of the reactor core and coolers dominate the total mass in most cases, and increasing the height of the reactor core can help reduce the system mass by allowing a smaller core radius. The Pareto Frontier shows that at thermal efficiencies around 40% or below, the system mass is depressed by using large CPR, molecular weight of the mixture, height of the reactor core, and small recuperator effectiveness. Increasing the recuperator's effectiveness from about 90% is required for higher thermal efficiency design, but its mass increases dramatically and can surpass the mass share of other components. The current analysis can provide a guide for detailed system and component design.

1. Introduction

Nuclear energy is being reconsidered worldwide as a low-carbon and dispatchable energy source [1]. Some of the large-scale nuclear power plants showed a significant cost overrun and time delay during their construction. As a result, the nuclear sector is showing increasing interest towards Small Modular Reactors (SMRs) as an option to generate electricity and process heat. Some SMRs are designed to be transportable and produce a relatively low power output of up to 20 MWe that can operate for industrial facilities and off-grid remote locations. This subset of SMRs is called microreactors [2]. They are being designed by several companies. The coolant choice for reactor design mainly includes gas, liquid metal with heat pipe, and liquid metal, and power conversion options are mainly the Brayton cycle and Rankine cycle [1].

The micro heat pipe-cooled reactor houses the heat pipe in a monolithic core, and hence a compact system is achieved by omitting the main pipeline, circulating pump, and auxiliary equipment [3]. A heat pipe microreactor, eVinci^{TM} is being developed by the Westinghouse Electric Company [4], and a 1.5 MWe microreactor, Aurora with heat pipe configuration is being designed by Oklo Inc. company [5]. However, this kind of reactor design is more mature at the kilowatt level than at the megawatt level [3]. Another common design option is to adopt a gas-cooled reactor with a direct closed Brayton cycle (CBC), in which the working fluid of the CBC is also the coolant of the reactor. It is beneficial for the compactness of the structure by eliminating the intermediate heat exchanger. Holos Generator [6], Xe-Mobile [7], MMR [8], and KAIST Micro Modular Reactor [9] all adopt this technological roadmap, except for the power conversion of MMR being Rankine cycle.

Helium, He-Xe mixture, and supercritical Carbon Dioxide (s-CO_2) are typical coolants for gas-cooled reactor concepts. Each has its advantages and disadvantages for being coolant as well as CBC working fluid. Helium as inert gas has good heat transfer and low-pressure loss but requires higher TIT and more turbomachinery stages. Helium CBC is one of the most mature of closed-cycle gas turbines [10]. The helium gas-cooled reactor has been intensively developed in China and is under

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Nomenclature

Latin symbols

A	heat transfer area (m^2)
c_p	specific heat capacity at constant pressure (J/kg/K)
C	capacitance rate (J/s/K)
D	diameter (m)
f	friction factor
h	specific enthalpy (J/kg) or convective heat transfer coefficient ($W/k/m^2$)
H	height of the reactor core (m)
k_{eff}	effective multiplication factor
$\dot{m}$	mass flow rate (kg/s)
m	mass (kg)
M	molecular weight of the He-Xe mixture (g/mole)
n	Rotation speed (rpm)
l	tube length (m)
N	node number
P	pressure (Pa)
q	average linear power density (W/m)
Q	heat (W)
R	thermal resistance ($W/K/m^2$) or radius of the reactor core (m)
T	temperature (K)
u	velocity (m/s)
UA	conductance (W/K)
z	location (m)

Greek symbols

η	efficiency
ε	effectiveness of the heat exchanger
κ	thermal conductivity (W/m/K)
ρ	density (kg/m^3)
λ	fluid thermal conductivity (W/m/K)
ζ	specific mass (kg/m^2)
ψ	head coefficient

Subscripts

allow	allowable
comp	compressor
core	reactor core
cold	cold side
ch	coolant channel
e	electrical
f	fuel
hot	hot side
i	element or node number
in	inlet
isen	isentropic
lv	latent heat of vaporization
ma	matrix
max	maximum
min	minimum
out	outlet
p	pinch point
pre	pre-cooler
rpc	recuperator
sat	saturation
sf	surface
turb	turbine
turbo	turbo-machines
tube	high-pressure tube
th	thermal
wt	water

Abbreviations

CBC	closed Brayton cycle
GT	gas turbine
He-Xe mixture	Helium-Xenon mixture
HP	high-pressure compressor / compressor 2
HX	heat exchanger
LP	low-pressure compressor / compressor 1
MSTE	micro shell and tube heat exchanger
Nu	Nusselt number
NTU	number of transfer units
RPM	Rotation per minute
s-CO_2	supercritical carbon dioxide
SMRs	small modular reactors
TIT	turbine inlet temperature

demonstration operation [12], though it uses a steam Rankine cycle. The large number of turbomachinery stages is not conducive to the application in nuclear microreactor systems. s-CO_2 has good efficiencies at moderate turbine inlet temperature (TIT) and small turbo-machineries, but it is more corrosive than helium at high temperatures [10]. Studies show that the nickel-base alloy has a shorter creep rupture lives in the s-CO_2 environment compared to those in air and atmospheric CO_2 [13], and increases in pressure will increase the oxidation and carburization rate, thus accelerating corrosion [14]. Historically, NASA considered He-Xe as the working fluid in their CBC power systems [15]. A more recent fluid trade study showed that He-Xe has the best efficiency and lowest heat exchanger volume among He-Xe, argon, and CO_2 working fluids, but it suggested the use of CO_2 due to the lower cost [11]. The He-Xe mixture retains the inertness and excellent heat transfer coefficient of pure helium, and it is proposed to reduce the size and mass of the turbomachinery for nuclear microreactors. By adding a large molecular weight gas of xenon, the enthalpy rise in the turbo-machine unit is decreased, hence reducing the average radius of the blades and the number of stages for the same rotation speed and aerodynamic loading [16]. Moreover, a performance evaluation on radial turbines shows that the turbines with He-Xe mixtures, Ar, and air share a pretty similar tendency of efficiency line and the values differ little [17]. Relevant studies on the mass and size optimization of the CBC systems are listed in Table 1.

Besides nuclear power, the CBC is also gaining attention in the field of concentrated solar power. Eldean et al. [25] present a performance study using different working fluids for the Concentrated Solar Gas Engine (Stirling or Brayton), and it shows that C_2H_2 achieves remarkable results. However, CO_2 is considered for both cycles operation putting in consideration the gas flammability and safety issues. Liu et al. [26] propose a CBC and organic Rankine Cycle (ORC) combined power generation system and explore the operation performance of different working fluid combinations during the lunar day process. Results show that the combination of He-Xe and toluene has the best performance among many working medium combinations. The performance of a He-Xe Brayton cycle coupled to the solar collector for a lunar base energy system is also assessed by Liu et al. [27]. It is found that as the helium mole fraction increases, the thermal efficiency and Bratyon-Rankine rotating unit mass increase, and adjusting the maximum operating pressure is useful for improving the system operation performance. The He-Xe is chosen in the current study given that it avoids corrosion problems compared with s-CO_2, and achieves smaller turbomachinery compared with pure helium. Moreover, it doesn't require operation pressure as high as 20 MPa for s-CO_2.

Table 1
Relevant studies on mass and size optimization of the CBCs.

Ref.	Year	Reactor type	Energy conversion system	Main findings
[18]	2014	Water-cooled reactor	s-CO_2 recompression cycle	The size of the turbomachinery can be ignored when compared to that of the heat exchanger, and newly proposed multiple shafts can increase flexibility and improve safety.
[19]	2015	Heat pipe-cooled reactor	He-Xe simple recuperation cycle	The increase in radiator panel area provides the highest potential to increase the power system efficiency, and there is a maximum system efficiency for each total heat exchanger mass.
[20]	2019	Gas-cooled reactor	s-CO_2 simple recuperation cycle	Radiator mass dominates the total mass, and a working fluid with a higher critical temperature may be more suitable given that typical advantages of an s-CO_2 system are not realized.
[21]	2020	Gas-cooled reactor	s-CO_2 Brayton cycle	The recompression cycle performs most outstanding in terms of total product unit cost among the studied cycles when the effectiveness of the heat exchanger exceeds 0.86.
[22]	2020	Gas-cooled reactor	He-Xe simple recuperation cycle	An increase of 4 % of the turbine inlet temperature from 1150 K results in a decrease of 6 % of the system mass, and the regenerator heat effectiveness is a key design parameter to minimize the system mass.
[23]	2022	Gas-cooled reactor	N_2O-He recompression cycle	The proposed novel working fluid has superior performance compared with N_2O and CO_2, and optimized thermal efficiency and Brayton rotating unit mass are 42.67 % and 3584.8 kg.
[24]	2022	Lead-cooled reactor	s-CO_2 recompression cycle	Increased turbine inlet temperature improves the thermoeconomic but augments the system volume, and the best trade-off result of three-objective optimization is a thermal efficiency of 42.14 %, a levelized cost of 50.3 $\$\bullet MWh^{-1}$, and a system volume of 4.43 m^3.

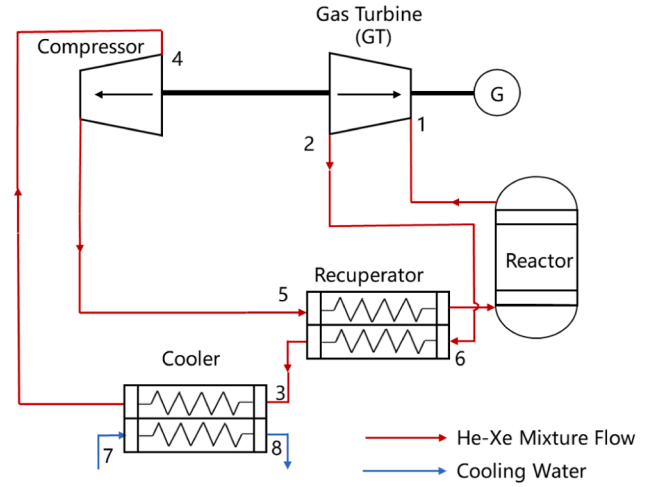
Most thermal efficiency and mass optimization of the He-Xe Brayton cycle is for space nuclear reactors, where heat exchangers or radiators comprise a sizeable percentage of the system mass [22]. For space nuclear reactors, the neutronic limitations cause the reactor size to be relatively unaffected by the power level [20]. Thus, the reactor mass is set as a constant value [202223].

However, for land-based nuclear microreactors, the reactor mass could account for a larger portion of the system mass considering a large thermal power. By coupling the thermodynamic calculation of the CBC, it is possible to search a minimum core radius according to the thermal limitations. The main contribution of this paper is to optimize the system mass and thermal efficiency of a He-Xe gas-cooled microreactor by considering the minimum core radius under different design conditions. To achieve this goal, component models with mass correlations are developed for the heat exchanger and turbomachinery. A 1D reactor model is established and coupled to the direct CBC to evaluate the minimal mass of the reactor core. A parametric study of the design parameters including turbine inlet temperature, compressor pressure ratio, minimum operating pressure, mixture molecular weight, reactor core height, and recuperator effectiveness on the thermal efficiency and system mass is performed. Finally, optimization of key design parameters concerning the system mass and thermal efficiency is conducted.

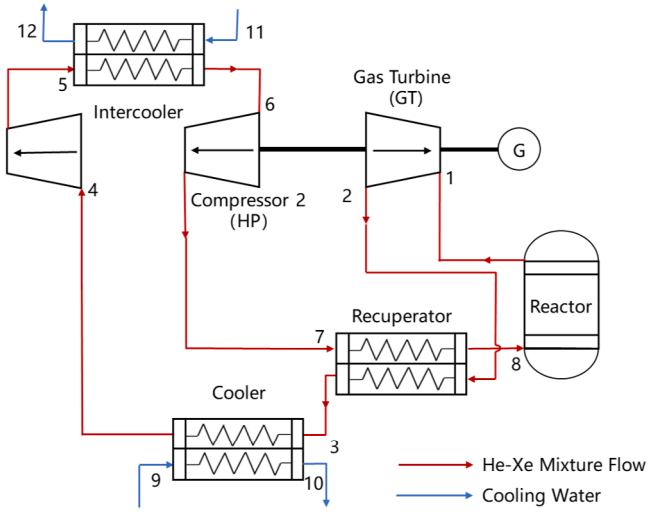
2. Systems description

As shown in Fig. 1, two typical cycle layouts for the He-Xe mixture are considered in the current work, where the gas-cooled nuclear reactor is directly connected to the CBC. The simple regenerative cycle achieves a compact layout with the least components. The intercooling cycle divides the compression into two stages and introduces an intercooler between stages to reduce compressor work and improve thermal efficiency. The turbo-machines use a single-stage centrifugal compressor, and a single-stage radial turbine on the same shaft. The micro shell and tube heat exchanger (MSTE) is used for the recuperators and coolers. It is a relatively new concept of compact heat exchanger, which could have a surface area density of 4500 m^2/m^3 [28]. Moreover, baffles are not typically required in MSTEs since the micro-sized tube and spacing can offer enough heat transfer without creating cross-flow.

The prismatic gas-cooled reactor concept uses a hexagonal assembly



a. Simple regenerative cycle (Config. A)



b. Intercooilng cycle (Config. B)

Fig. 1. He-Xe Brayton cycle layouts.

design, which usually adopts a graphite-based block with coolant channels and fuel channels. A single assembly is shown in Fig. 2, which contains 19 fuel pins and 54 coolant channels. The fission heat released from the uranium carbide fuel is removed by the He-Xe coolant. Coolant channels made of high-pressure tubes, together with the fuel pins are housed in the graphite matrix. The melting points of the uranium carbide fuel and graphite moderator are very high, about 2780 K and 3872 K, respectively. Therefore, the maximal allowable temperature of the assembly is limited by the high-pressure tube made of Mo-TZM, and it is set as 1400 K for the current study [29]. Detailed geometry sizes of the assembly and reactor are listed in Table 2, which are based on the core design of the author's previous work [30].

3. System modeling

3.1. Cycle modeling methods

A Brayton cycle model with mass optimization capabilities is developed in Python. Models for the main components are constructed based on their thermodynamic process. The power cycle components models include the reactor, recuperator, compressor, turbine, and pre-cooler. The semi-empirical relations proposed by Tournier and El-Genk [31] are used to calculate the thermo-physical properties of the He-Xe mixture. Fig. 3 displays the variations of specific heat capacity and viscosity with the molecular weight. As the mole fraction of xenon increases, the specific heat capacity of the He-Xe mixture decreases rapidly, while the dynamic viscosity initially increases and then slightly decreases. The change in thermophysical properties of He-Xe mixtures will affect the heat transfer performance, pressure losses, and output power of the cycle.

Due to the large number of calculations required for parametric and optimization analyses, the one-dimensional model is used for the reactor core for the early design stage [32]. The one-dimensional discretization along the flow direction is shown in Fig. 4. Due to the irregular geometry of the graphite matrix, related heat conduction is calculated based on the process through a cylindrical wall with an equivalent area. The radial power distribution is assumed to be uniform, while the axial power is assumed to be a cosine distribution. Therefore, the calculation can be conducted for one single channel,

$$Q_{ch} = c_p \cdot \dot{m}_{ch} \cdot (T_{core,out} - T_{core,in}) \tag{1}$$

where the Q_{ch} is the heat per channel, c_p is the heat capacity of coolant, $\dot{m}_{ch}$ is the mass flow rate per channel, $T_{core,in}$ and $T_{core,out}$ is the inlet and outlet coolant temperature of the reactor core, respectively. The coolant channel is discretized into N elements, and for element i, the coolant temperature can be calculated as,

$$T_{core,i} = T_{core,in} + \frac{Q_{ch,i}}{\dot{m}_{ch} c_p} \tag{2}$$

where the $Q_{ch,i}$ is the sum of the heat removed from the core inlet to element i, and can be calculated through integration,

$$Q_{ch,i} = \int_{-0.5H}^{z_i} q_l \cdot \frac{\pi}{2} cos(\frac{\pi z}{L}) dz \tag{3}$$

where the $q_l = Q_{ch}/L$ is the linear power density, H is the length of the reactor core, $z_i = -0.5L + i \cdot (H/N)$ is the position of the element i. Thus, the surface temperature of the high-pressure tubes can be calculated,

$$T_{tube,i} = T_{core,i} + \frac{q_l \cdot \pi / 2 \cos(\pi z_i / L)}{A_{tube,i} h_{core,i}} \tag{4}$$

where $A_{tube,i}$ is the heat transfer area of element i, $h_{core,i}$ is the heat transfer coefficient, which can be calculated through Nusselt number,

$$Nu = \frac{h_{core} D_{tube,in}}{\lambda_{heXe}} \tag{5}$$

where $D_{tube,in}$ is the inner diameter of the high-pressure tube, λ_{hexe} is the thermal conductivity of the He-Xe coolant. The Nusselt number is calculated using the correlation proposed by Taylor [33].

Before the heat convection of the He-Xe coolant, the fission heat released by the fuel pin is first transferred to the matrix and high-pressure tubes by heat conduction. Temperatures of different materials can be calculated based on corresponding thermal resistances,

$$T_{f,i} = T_{sf,i} + \frac{q_l \cdot \pi / 2 cos(\pi z_i / L)}{R_{tube} + R_{ma} + R_f} \tag{6}$$

where $T_{f,i}$ the is the temperature of the fuel center, R_{tube}, R_{ma}, and R_f are thermal resistances of the high-pressure tube, matrix, and fuel.

The pressure drop in the core is calculated using the following equation,

$$P_{i+1} = P_i - \frac{f_i}{2} \frac{\delta l_i}{D_{tube}} \cdot \rho_i u_i^2 \tag{7}$$

where P_i is the pressure value of element i, f_i is the friction factor given by empirical correlations in reference [35], δl_i is the tube length, ρ_i is the fluid density, and u_i is the fluid velocity.

The recuperator and pre-cooler are also discretized into a set of sub-heat exchangers as displayed in Fig. 5. The pressure drops in these two heat exchangers are assumed to be constant. For the recuperator, the outlet temperatures are first calculated according to the definition of effectiveness. Because the heat capacity of He-Xe barely changes with the temperature and pressure, the effectiveness of the recuperator is calculated by

$$\epsilon = \frac{T_{rpc,hot,in} - T_{rpc,hot,out}}{T_{rpc,hot,in} - T_{rpc,cold,in}} \tag{8}$$

Combined with the energy conservation, the outlet temperatures on both sides of the recuperator can be obtained.

For the pre-cooler, the water is used to cool the He-Xe mixture, and the temperature difference at the pinch point is used as input. Therefore, the energy balance of the pre-cooler is given by

$$\dot{m}_{wt} h_{lv} = \dot{m} c_p (T_{pre,in} - T_{wt,sat} - \Delta T_p) \tag{9}$$

Table 2
Geometry sizes of the fuel assembly and reactor core [30].

Parameter	Value
Radius of fuel [cm]	0.75
Radius of the coolant channel [cm]	0.4
High-pressure tube thickness [cm]	0.05
Pitch [cm]	1.5
Apothem [cm]	6.75
Radius of Core [m]	0.5
Height of Core [m]	0.8

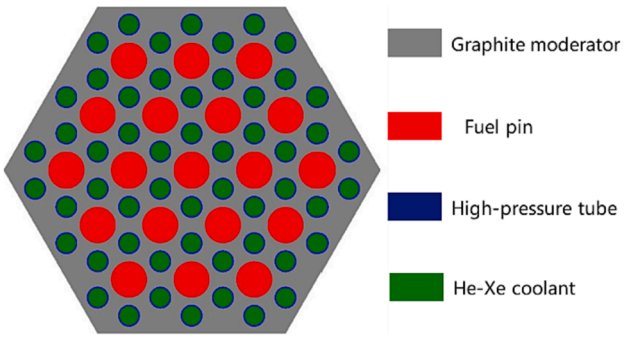


Fig. 2. Assembly geometry of the gas-cooled reactor.

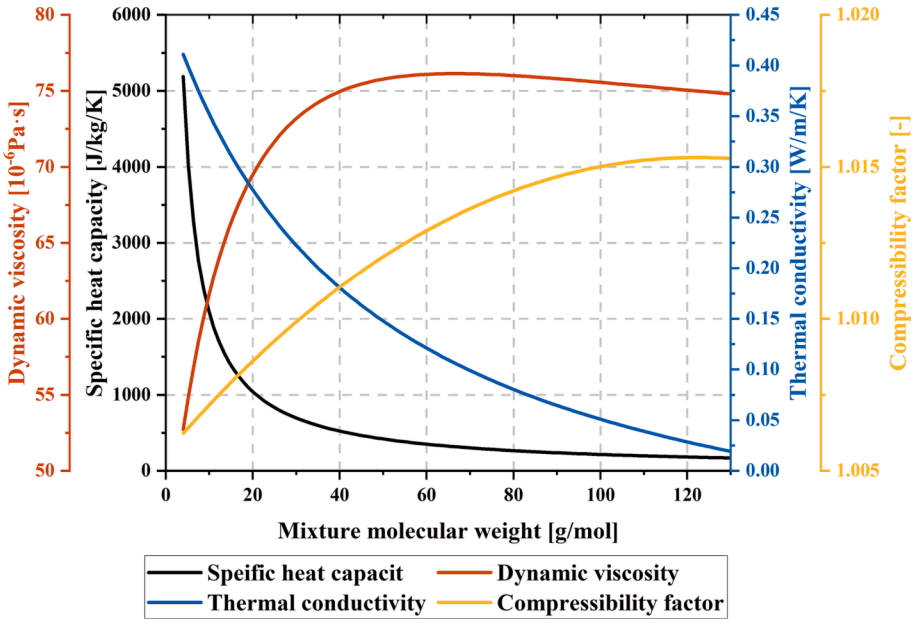


Fig. 3. The thermophysical properties of the He-Xe mixture.

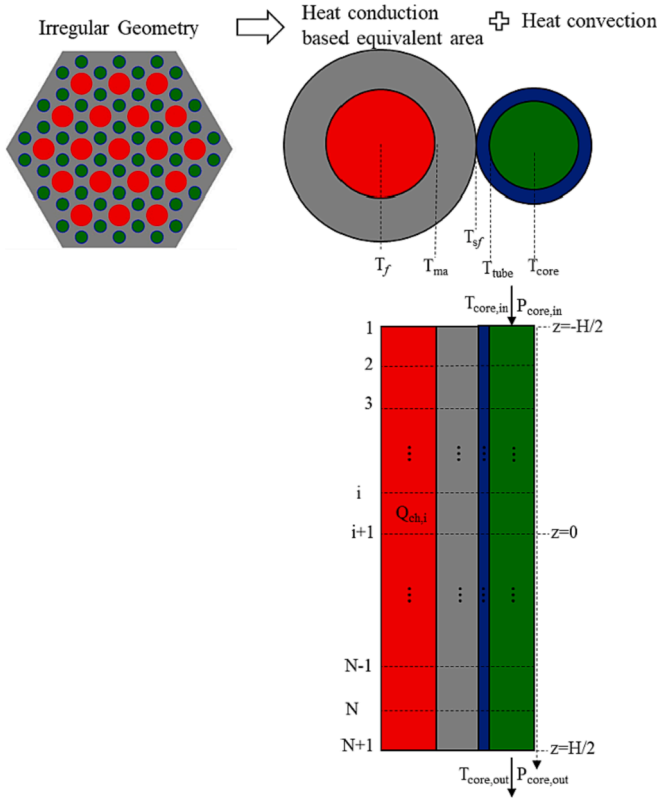


Fig. 4. One-dimensional discretization of the reactor model.

$$\dot{m}_{wt}(h_{wt,sat} - h_{wt,in}) = \dot{m}c_p(T_{wt,sat} + \Delta T_p - T_{pre,out}) \tag{10}$$

where $\dot{m}_{wt}$ is the mass flow rate of the water flow, h_{lv} is the latent heat of vaporization of water, $T_{pre,in}$ and $T_{pre,out}$ is the inlet and outlet temperature of He-Xe flow, $T_{wt,sat}$ is the saturated temperature of the water, ΔT_p is the temperature difference at pinch point, $h_{wt,sat}$ and $h_{wt,in}$ is the saturated and inlet enthalpy of the water flow.

It is assumed that each sub-heat exchanger has an equal heat duty. Therefore, the outlet temperature of sub-heat exchanger i can be calculated by

$$\frac{Q}{N} = \dot{m}c_p(T_i - T_{i+1}) \tag{11}$$

Once the temperature distribution is obtained, the conductance of each sub-heat exchanger UA_i can be calculated using the NTU-effectiveness method, in which the fluid properties of each sub heat-exchanger are calculated based on the average temperature and pressure [34],

$$UA_i = NTU \cdot C_{min} \tag{12}$$

$$NTU = \begin{cases} \dfrac{\log(\dfrac{1-\epsilon \cdot C_R}{1-\epsilon})}{1-C_R}, if\ C_R \neq 1 \\ \dfrac{\epsilon}{1-\epsilon}, otherwise \end{cases} \tag{13}$$

$$C_R = \frac{C_{\min}}{C_{max}} \tag{14}$$

where UA_i is the conductance, NTU is a dimensionless number of transfer units for sub-heat exchanger i, C_{min} and $C_{\max}$ is the minimal and maximal capacitance rate. Finally, the total conductance is summed by

$$UA = \sum_{i=1}^{n} UA_i \tag{15}$$

The compressor outlet temperature is calculated based on the isentropic efficiency,

$$h_{comp,out} = h_{comp,in} + \frac{h_{comp,out,isen} - h_{comp,in}}{\eta_{comp}} \tag{16}$$

where the $h_{comp,in}$ and $h_{comp,out}$ is the specific enthalpy at the inlet and outlet of the compressor, $h_{comp,out,isen}$ is the specific enthalpy at the outlet of the compressor if the fluid is isentropically processed to the outlet pressure. Similarly, the turbine outlet temperature can be calculated by

$$h_{turb,out} = h_{turb,in} - (h_{turb,in} - h_{turb,out,isen})\eta_{turb} \tag{17}$$

Therefore, the thermal efficiency of the cycle is defined as

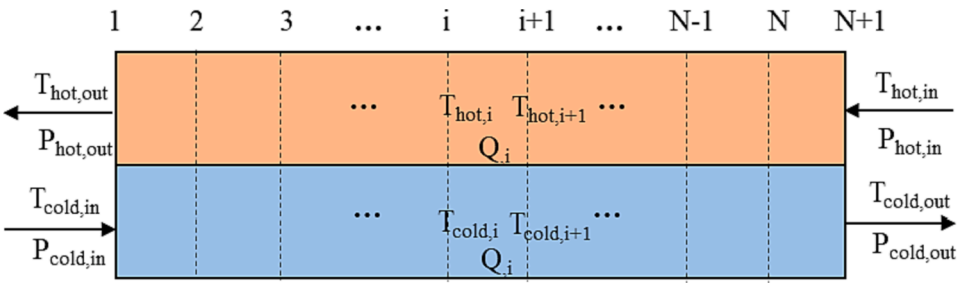


Fig. 5. One-dimensional discretization of the heat exchanger model.

$$\eta_{th} = \frac{W_{turb} - W_{comp}}{Q_{core}} \tag{18}$$

where Q_{core} is the heat input from the reactor core, $W_{turb} = \dot{m}(h_{turb,in} - h_{turb,out})$ is the turbine work, W_{comp} is the compressor consumed work.

The component models are integrated to create the simple regenerative cycle and intercooling cycle. The calculation flowchart of the simple regenerative cycle is displayed in Fig. 6. The input design parameters mainly include target electric power output, efficiencies of turbo-machines and recuperator, compression ratio, temperature difference at pinch point for pre-cooler, geometric parameters of the reactor core, helium mole fraction of the He-Xe mixture. By assuming a pressure loss of each component, the pressure distribution along the loop is first calculated. Then, outlet temperatures of the compressor and turbine are calculated and passed to the recuperator model. The recuperator model determines the inlet temperatures of the reactor and pre-cooler. In the reactor model, axial temperature distributions of different materials are obtained. In addition, the thermal power and pressure loss are also calculated. In the pre-cooler model, the outlet temperature of He-Xe flow and the required mass flow rate of the cooling water are determined. Next, if the electric power output does not match the target one, the mass flow rate in the cycle will be updated, and the above calculation will be repeated. Finally, the radius of the reactor core is reduced until the maximal temperature of the pressure tube reaches the design safety value, 1400 K. By this means, it is possible to search the minimum allowable value of core radius under a specified electric power output.

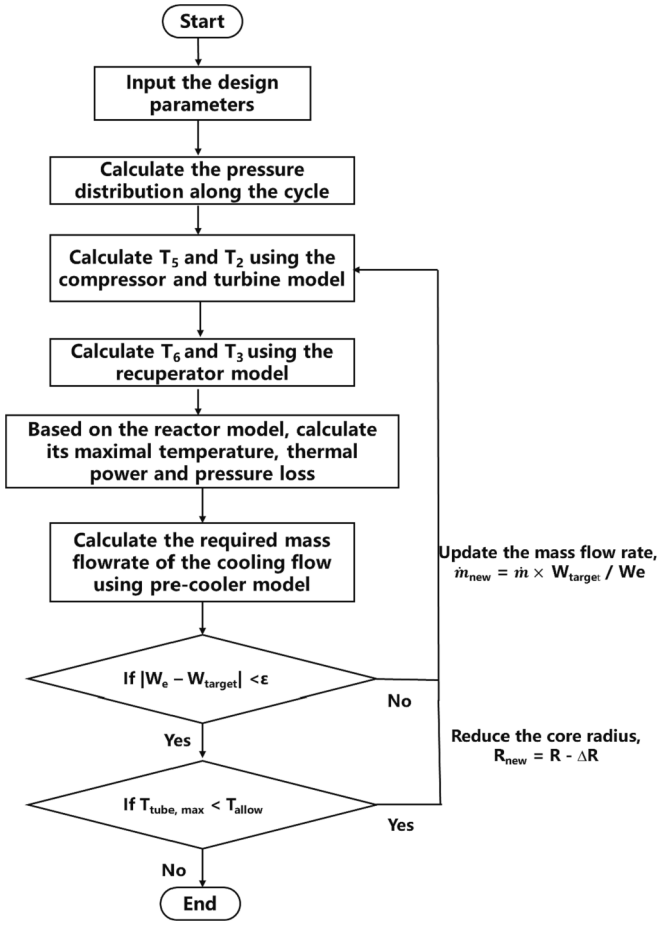


Fig. 6. The flowchart of the iteration process for the simple regenerative cycle.

3.2. Mass correlation

It would benefit from the mass optimization for the land-based, mobile reactor considering the limitation of transportation capacity. The system mass optimization considers the masses of the reactor core, heat exchangers, and turbo-machines. The mass of the working fluid, piping system, and other auxiliary components are neglected. The mass of the reactor core is directly calculated based on geometric parameters and material densities. In terms of the recuperator and pre-cooler, a mass correlation based on the provided data from industry patterner is used to calculate the mass of the micro shell and tube heat exchanger [20]. In this correlation, the conductance is used to be correlated directly to its mass,

$$m_{HX} = \xi_{HX} \cdot UA + 16.45 \tag{19}$$

where ξ_{HX} is the specific mass of the heat exchanger, 2.11 $kg{\cdot}K/kW$, and the header mass is not included when using this correlation.

The energy conversion component includes a single-stage centrifugal compressor, a single-stage radial turbine, and a smooth-rotor alternator on the same shaft. The mass of the turbo-machine is computed as [3637],

$$m_{turbo} = \xi_{turbo} \cdot D^2 \tag{20}$$

where ξ_{turbo} is the specific mass of the turbo-machine, 5360 kg/m^2. D is the diameter, which can be calculated through the definition of head coefficient,

$$D = \frac{\sqrt{c_p \cdot \frac{\Delta T}{\psi}}}{\pi \cdot n} \tag{21}$$

where ψ is the head coefficient, c_p is the specific heat capacity, ΔT is the temperature difference between the inlet and outlet, n and is the rotating speed.

4. Results and discussion

4.1. Validation

In order to validate the model of the He-Xe Brayton cycle presented in this study, a simple regenerative cycle layout referenced from the literature [38] and an intercooling cycle layout from the literature [39]

are established. Compared with the intercooling cycle in Fig. 1(b), the referenced one is coupled with two Organic Rankine Cycles (ORC) by introducing two evaporators before the precooler and intercooler, respectively. The validation calculation is conducted using the same assumptions as the reference literature. Moreover, the calculation takes the compressor pressure ratio, compressor inlet temperature, pressure, turbine inlet temperature, and pressure drops of components as inputs. The thermal efficiencies as well as temperature distributions are calculated and compared. Temperature distributions of both the intercooling cycle and simple regenerative cycle agree well as listed in Table 3 and Table 4. The slight deviations could be caused by the differences in the thermodynamic properties database and the measurement accuracy of the experiment. The comparison of the thermal efficiency is summarized in Table 5.

4.2. Parametric study

In this section, parametric studies are conducted to investigate the effects of different design parameters on the system mass and thermal efficiencies. Input values of basic design parameters are listed in Table 6. The variables considered in the parametric study include TIT, compressor pressure ratio (CPR), compressor inlet pressure (CIP), and the molecular weight of the He-Xe mixture. The effects of one parameter are examined by keeping the others constant. The power output is fixed at 8 MWe, which is achieved by adjusting the mass flow rate in the circuit under different design parameters.

Fig. 7 displays the effects of the turbine inlet temperature. As the TIT increases, thermal efficiencies of the intercooling cycle and simple regenerative cycle increase, and the system mass decreases. With the same loop flow rate, an increase in TIT implies a growth in the turbine work as well as output power. The calculated output power is fixed at 8 MWe in the current model, which means the searched flow rate of the loop is smaller. The reactor thermal power is affected by both the TIT and mass flow rate. The decrease in flow rate has a greater impact on the reactor power than the increase in TIT. Therefore, the thermal power of the reactor decreases, resulting in an improvement in system efficiency. Moreover, the reduced mass flow rate leads to a decrease in capacitance rates and conductances of the recuperator and cooler, thereby reducing its mass. The reactor mass remains unchanged with fixed dimensions, which will be discussed later. The turbine mass increases as the enthalpy change grows. The decrease in heat exchanger mass dominates, resulting in a reduction in system mass. The intercooling cycle can achieve a higher thermal efficiency but its system mass is also bigger due to the introduction of intercooler and compressor 2 as shown in Fig. 7 (b).

Table 3
Validation results of temperature distributions for the intercooling cycle.

Section conditions		Ref. [39]		This work	
	Fluid	P [kPa]	T [℃]	P [kPa]	T [℃]
Turbine inlet	Helium	7930	850	7930	850
Recuperator hot side inlet	Helium	2630	475.5	2630	475.4
Evaporator 1 hot side inlet	Helium	2580	157.7	2580	161.2
Precooler inlet	Helium	2540	73.31	2540	72.4
LP compressor inlet	Helium	2500	29.83	2500	29.83
Evaporator 2 hot side inlet	Helium	4503	119.7	4503	119.7
Intercooler inlet	Helium	4463	69.77	4463	70.0
HP compressor inlet	Helium	4423	29.48	4423	29.7
Recuperator cold side inlet	Helium	8110	122.4	8110	122.8
Reactor core inlet	Helium	8030	440.1	8030	437.3
Pump inlet (ORC 1)	R123	154.7	40	154.5	40.0
Evaporator 1 inlet (ORC 1)	R123	764.3	40.32	763.0	40.3
Turbine 1 inlet (ORC 1)	R123	764.3	98.7	763.0	98.7
Condenser 1 inlet (ORC 1)	R123	154.7	57.09	154.5	56.4
Pump inlet (ORC 2)	R123	154.7	40	154.5	40.0
Evaporator 2 inlet (ORC 2)	R123	477.1	40.18	476.4	40.2
Turbine 2 inlet (ORC 2)	R123	477.1	78.96	476.4	79.0
Condenser 2 inlet (ORC 2)	R123	154.7	51.06	154.5	50.7

Table 4
Validation results of temperature distributions for the simple cycle.

Section conditions	Ref. [38]		This work	
	P [kPa]	T [℃]	P [kPa]	T [℃]
Compressor inlet	7780	33.2	7780	33.2
Recuperator cold side inlet	20,000	74.3	20,000	74.3
Heater inlet	19,900	269.1	19,900	269.1
Turbine inlet	19,800	500	19,800	500.0
Recuperator hot side inlet	8030	401.5	8030	401.5
Precooler inlet	7930	116.2	7930	116.2

Table 5
Validation results for the system efficiency.

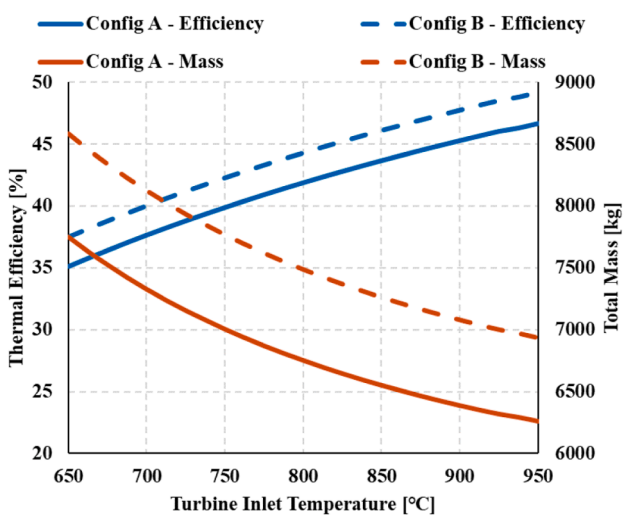
	System efficiency η_{th}[%]
Ref. [39]	49.80
This work	49.55
Ref. [38]	25.6
This work	25.57

Table 6
Basic design parameters of the simple regenerative and intercooling cycle.

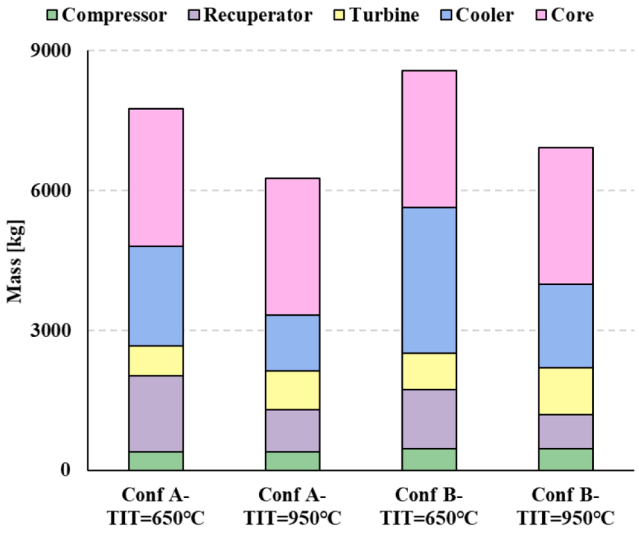
Parameter	Symbol	Value
Power output [W]	W_{out}	8×10^6
Cooling water temperature [℃]	T_0	25
Cooling water Pressure [kPa]	P_0	101.325
Compressor inlet temperature [℃]	$T_{c,in}$	35
Compressor inlet pressure [kPa]	$P_{c,in}$	2500
Compressor pressure ratio [-]	Π	2.4/3.0
Mixture molecular weight [g/mol]	M_{mix}	29.5
Turbine inlet temperature [℃]	$T_{t,in}$	850
Effectiveness of recuperator [-]	η_{rec}	90 %
Isentropic efficiency of turbine [-][40]	η_t	93 %
Isentropic efficiency of compressor [-][40]	η_c	89 %
Pressure loss in reactor core [kPa] [39]	ΔP_{core}	100
Pressure loss in the HP side of the recuperator [kPa] [39]	$\Delta P_{rec,hp}$	80
Pressure loss in the LP side of the recuperator [kPa] [39]	$\Delta P_{rec,lp}$	50
Pressure loss in other components [kPa] [39]	ΔP	40
Rotating speed [RPM][40]	n	28,800
Head coefficient [-][37]	Ψ	0.66
Temperature difference at pinch point [℃] [39]	ΔT_p	5

When the TIT is 1123 K, the intercooling cycle has a thermal efficiency of 46.12 % and a system mass of 7259.78 kg, while the simple regenerative cycle has a thermal efficiency of 43.66 % and a system mass of 6551.97 kg. However, the recuperator mass of the intercooling cycle is smaller than that of the simple regenerative cycle. This is also because the mass flow rate of the intercooling cycle is smaller.

Fig. 8 illustrates the effects of the compressor pressure ratio. As shown in Fig. 8(a), when the CPR is increased, the thermal efficiency first increases and then decreases, and the decrease is greater for the simple regenerative cycle. Because the expansion ratio which influences the turbine work, is determined by the CPR and pressure losses of heat exchangers as well as the reactor. The optimal CPR for thermal efficiency is related to the pressure losses of components in the cycle. Under the current pressure losses of components, the maximal thermal efficiency of the simple regenerative cycle is 43.76 % at the CPR of 2.27. For the intercooling cycle, it is 46.29 % at the CPR of 2.68. Further increasing the CPR leads to the growth of compressor work exceeding that of the turbine, and hence the thermal efficiency begins to drop. The growth of compression work in the intercooling cycle is slower since the inlet temperature of compressor 2 is cooled first. Consequently, its thermal efficiency drops more slowly. Concerning the system mass, increasing the CPR results in a larger enthalpy change of the compressor and turbine, which leads to a growth in their masses as shown in Fig. 8 (b). The masses of the heat exchanger are determined by the

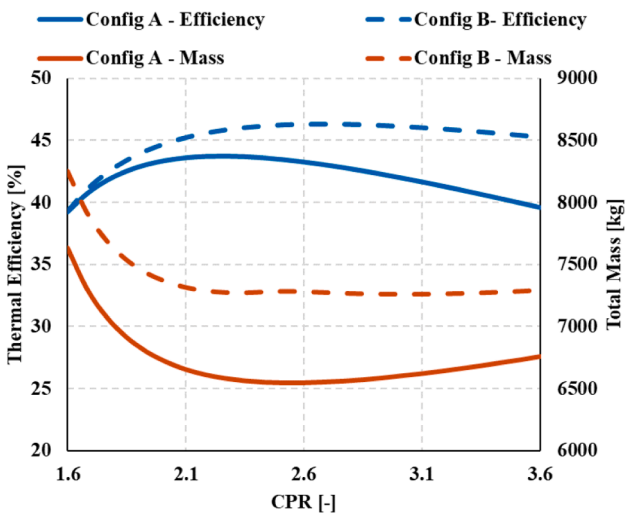


(a) efficiency & mass

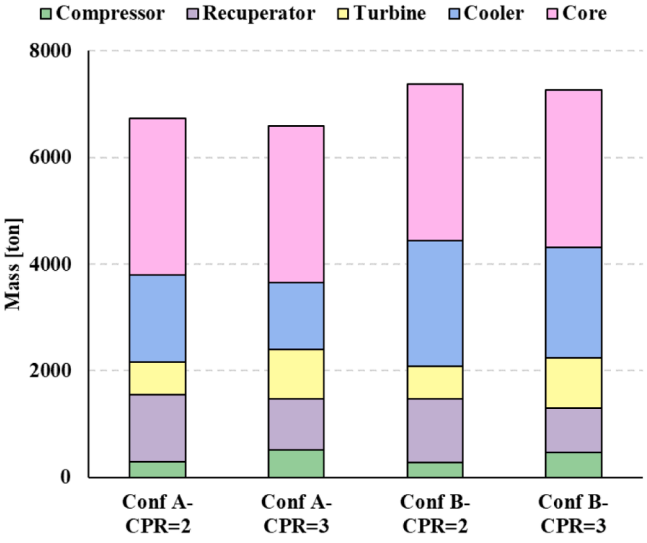


(b) mass composition

Fig. 7. Effects of the turbine inlet temperature.



(a) efficiency & mass



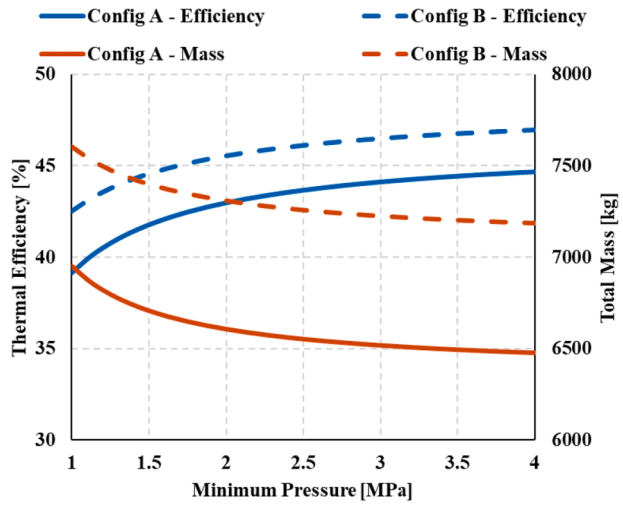
(b) mass composition

Fig. 8. Effects of the compressor pressure ratio.

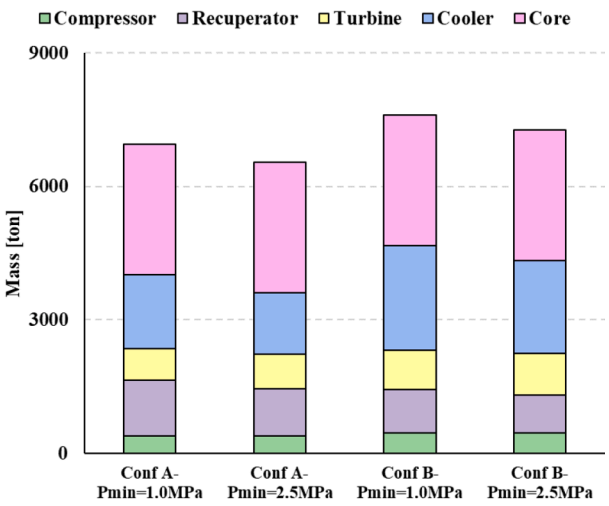
conductance. Because the specific heat capacity of the He-Xe mixture barely changes with temperature and pressure, conductance values are mainly related to the cycle flow rate. As the CPR increases, the inlet temperature on the recuperator cold inside grows. At the same time, the inlet temperature on the recuperator hot side drops due to the increase in expansion ratio. Both lead to a deterioration in the heat transfer capacity of the recuperator. Thus, the reactor inlet temperature drops and the searched mass flow rate continues to reduce for an 8 MWe design. As a result, the masses of the recuperator and cooler decrease. At first, the mass reduction of the heat exchanger dominates, and hence the system mass decreases until the growth of the turbomachinery mass dominates. The minimal mass of the simple regenerative cycle is 6543.8 kg at the CPR of 2.52, and for the intercooling cycle, it is 7260.0 kg at the CPR of 3.02.

Fig. 9 displays the effects of the compressor inlet pressure. The rise in cycle minimum pressure enlarges the expansion ratio in the turbine, thereby improving the cycle efficiency as shown in Fig. 9(a). The system mass decreases as the CIP increases, which mainly depends on the mass of the cooler and recuperator as displayed in Fig. 9(b). It is similar to the situation where the TIT is increased. When the expansion ratio in the turbine is enlarged at a higher CIP, the thermal efficiency is increased. A smaller mass flow rate is obtained for an 8 MWe design, and hence masses of the recuperator and cooler reduce.

The effects of the mixture's molecular weight are illustrated in Fig. 10. Both the cycle efficiency and system mass decline as the mixture's molecular weight increases. The pressure loss in the reactor grows as the mole fraction of xenon gas increases. As a result, the turbine work as well as the thermal efficiency reduces. The descent rate of system

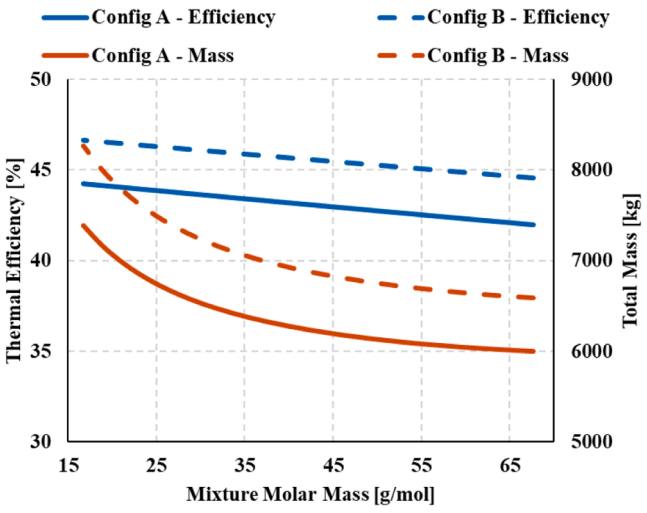


(a) efficiency & mass

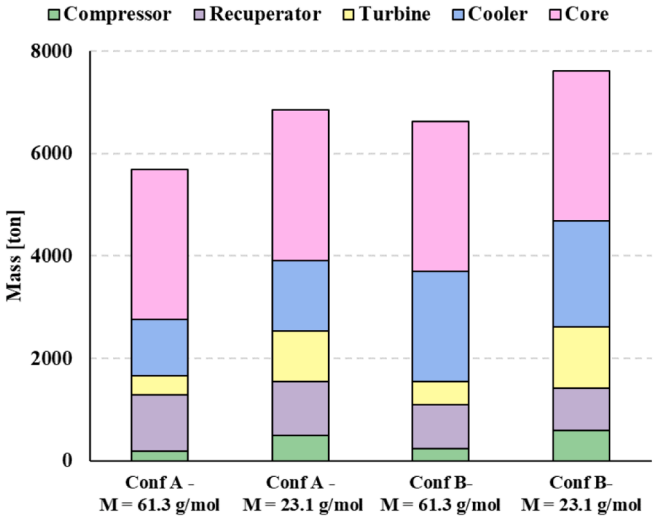


(b) mass composition

Fig. 9. Effects of the compressor inlet pressure.



(a) efficiency & mass



(b) mass composition

Fig. 10. Effects of the mixture's molecular weight.

mass is larger when the mixture's molecular weight is less than 35 g/mol. The molar heat capacity of noble gases and binary mixtures at the inlet temperature of the compressor is nearly constant [16]. As a result, the specific heat capacity is inversely proportional to the molecular weight of the He-Xe mixture, and it drops significantly when the molecular weight is less than 35 g/mol. Therefore, the diameters and masses of the compressor and turbine reduce when the mixture's molecular weight increases as shown in Fig. 10(b).

There are two compressors in the intercooling cycle, and Fig. 11 shows their effects individually, in which the CPR in the x-axis represents the overall pressure ratio achieved by the two compressors. For the same pressure ratio of LP, when increasing the CPR by raising the pressure ratio of HP, the system mass firstly decreases, and then slightly increases. This is similar to the situation when the CPR is increased in a simple regenerative cycle. The temperature difference between the hot and cold inlet of the recuperator narrows. The reactor inlet temperature drops and the searched mass flow rate continues to reduce for an 8 MWe design. As a result, the masses of recuperator and coolers decrease. The system mass reduces until the growth of the turbomachinery mass dominates. On the other hand, lowering the pressure ratio of LP will help reduce the system mass when the overall CPR remains constant. This is because the thermal efficiency variation is relatively small when the overall CPR is fixed, and thus changes in the cycle flow rate, as well as masses of the recuperator and precooler are small. However, the intercooler mass significantly increases as the outlet temperature of LP grows. Furthermore, since the specific heat capacity of the He-Xe barely changes with pressure and temperature, the power consumption of the compressors is related to the outlet temperatures of the two compressors for a certain CPR. With the pressure loss in the intercooler neglected, using a compression ratio of $\sqrt{CPR}$for both compressors can obtain the lowest outlet temperature and the highest thermal efficiency. Therefore, it can be seen that the impact of the compression ratio of LP on the

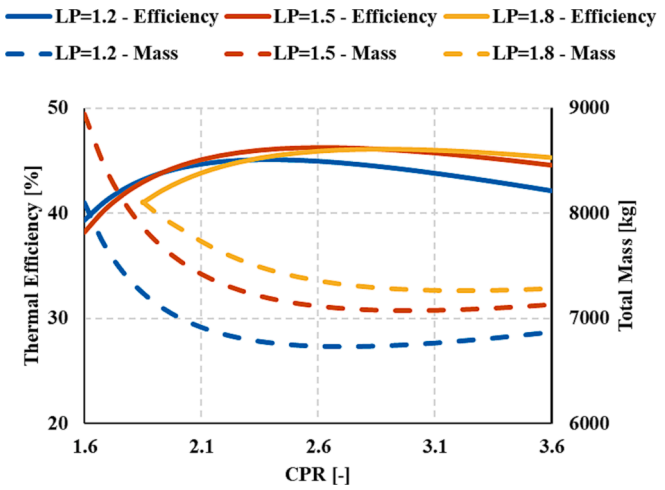


Fig. 11. Effects of compressor pressor ratios for HP and LP.

thermal efficiency is nonlinear.

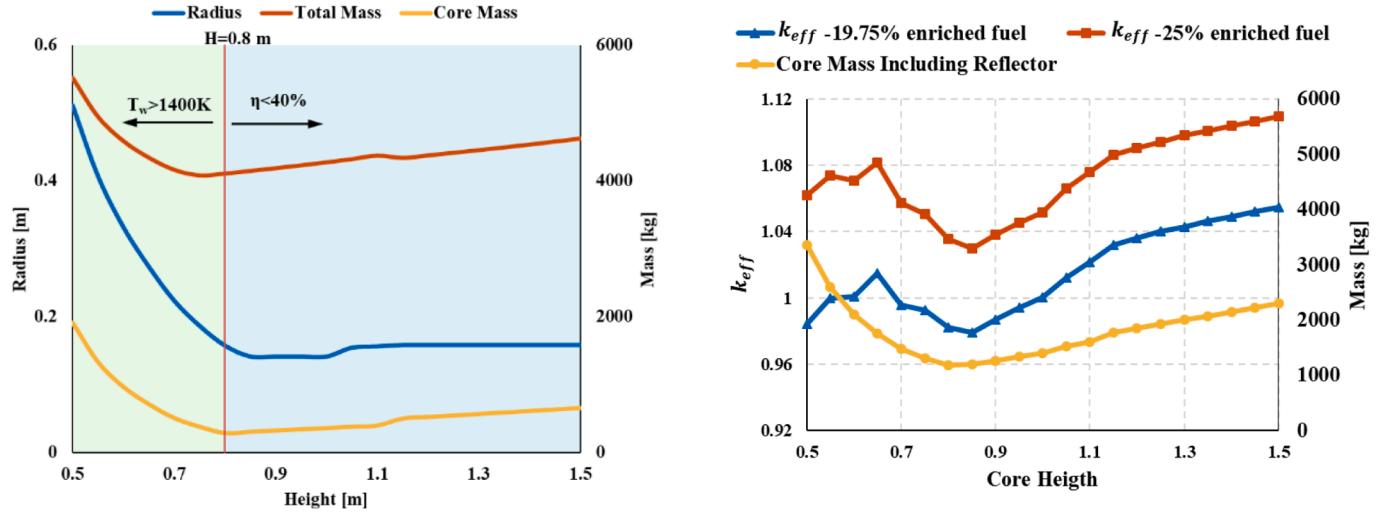
The geometric dimensions of the reactor core are fixed in the above analysis. Next, the effects of reactor dimensions on system mass are discussed. As the simple regenerative cycle has a smaller system mass, it is used in the following analysis. The current model considers two constraints regarding the size of the core. The first one is that the maximal temperature of the high-pressure tube in the coolant channel cannot exceed the design value of 1400 K, and the second is that the pressure drop in the reactor core cannot exceed a certain value which results in a thermal efficiency smaller than 40 %.

Under these two constraints, the minimal core radii of each height are searched and plotted in Fig. 12(a). When the core height is enlarged, smaller core radii can be searched. This is because the linear power density of each coolant channel is reduced, and fewer coolant channels are allowed while ensuring that the maximal tube temperature is below 1400 K. The corresponding core radius can be reduced. However, fewer coolant channels make the pressure drop in the reactor increase greatly. As the core height further grows, the minimal core radius increases slightly to reduce the pressure drop and keep the cycle efficiency above 40 %. Such transformation happens at the core height of around 0.8 m and the corresponding minimum core weight is 293.8 kg. For the total system mass, the minimal value is 4082.6 kg at a height of 0.75 m. Although the core mass is increased at a height of 0.75 m, the reduction of the pressure loss in the reactor helps improve the thermal efficiency leading to a greater decrease in the masses of other components.

It should be noted that the dimensions of the reactor core should also meet the neutron critical requirement to keep sustainable fission reactions besides temperature limits. Fig. 12(b) illustrates the variations of multiplication factors (k_{eff}). The results are calculated by using OpenMC [41], and the largest error of k_{eff} is below 0.00010. Moreover, a 30 cm thick radial reflector and a 10 cm thick axial reflector are adopted to reduce neutron leakage, which is made of pure beryllium. At the core height of 0.7 ~ 0.95 m, the core mass is relatively small. However, these designs cannot achieve criticality when the enrichment of ^{235}U is 19.75 %. This is caused by the high neutron leakage with small core radii. By further increasing the core height beyond 0.95 m, the core can achieve criticality, and the k_{eff} gradually grows due to the increasing core dimensions as well as fuel loading. However, the core mass is also increased at a faster rate with the existence of the reflectors. Comparatively, using large core heights with small radii can have smaller core masses and larger k_{eff}. When the enrichment is increased to 25 %, designs at all core heights have a k_{eff} greater than 1. Adjustments of the fuel diameter and reflector thickness are also options to help achieve criticality given that the uranium for civilian use should be limited to 20 %. In summary, the core mass calculated by the one-dimensional model acts as the base mass that satisfies the requirement of material temperature limits, additional masses including the reflectors may be required to achieve criticality.

Results of total mass with different mixture molecular weights are shown in Fig. 13. The total mass decreases with increasing molecular weight. For the molecular weight of 16.7 g/mol of which the helium mole fraction is 90 %, the total mass is 4853.2 kg at the core height of 0.85 m. For the molecular weight of 67.6 g/mol of which the helium mole fraction is 50 %, the total mass reduces to 3787.6 kg. Although the required core masses are bigger for larger molecular weights due to the growth of thermal power, the mass reductions of the compressor and turbine are dominant. However, the thermal efficiency is also reduced as the molecular weight grows. Under the current condition, the core height of 0.8 ~ 0.85 m is suitable for mixtures with different molecular weights as it brings the least total mass. Within this range of core height, the core mass is at its lowest level and high thermal efficiency ensures low masses of other components.

The effects of recuperator design on system mass and efficiency are



(a) system mass (b) critical analysis

Fig. 12. Effects of the reactor core dimensions on the system mass.

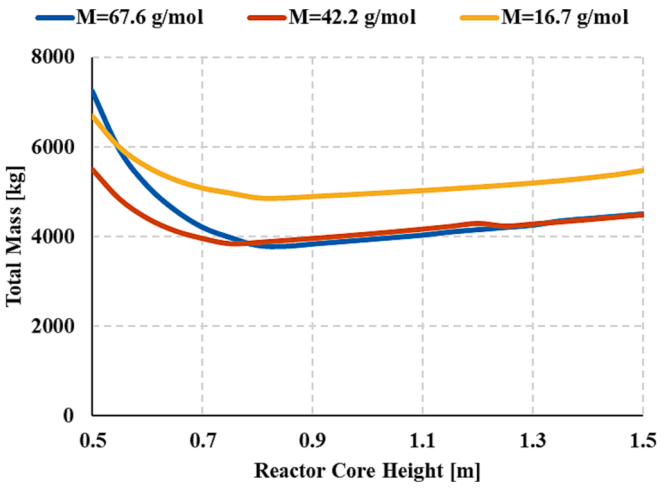


Fig. 13. Total mass variation with core height under different mixture molecular weights.

plotted in Fig. 14~Fig. 16. The recuperator conductance, effectiveness, and capacitance rate are interrelated. Fig. 14 demonstrates the effects of the recuperator effectiveness for different conductance values. For a fixed conductance value, the mass flow rate is inversely proportional to the recuperator effectiveness, which can be deduced from Equation (12) and (13). When the recuperator effectiveness is low, it results in a large mass flow rate and excessive pressure loss in the reactor core. Hence, the expansion ratio as well as the work of the turbine is small. As the mass flow rate reduces with increasing effectiveness, the pressure loss is lowered. The turbine work briefly increases due to the growing expansion ratio, and then gradually decreases. At the same time, the work of the compressor continues to decrease with constant CPR. Therefore, as displayed in Fig. 14(a), there is an optimal recuperator effectiveness where the power output is the maximum for a certain conductance. The optimal efficiency of the recuperator shifts towards larger values as the conductance grows. This is because a higher conductance results in larger recuperator effectiveness for the same range of mass flow rate. On the other hand, using a small conductance design with large recuperator efficiency can help reduce the total mass as shown in Fig. 14(b). However, such effects gradually weaken as the recuperator efficiency grows. The mass flow rate is proportional to the conductance under the same effectiveness, and hence a lower conductance results in a lower power level and system mass. As the recuperator effectiveness increases, the mass flow rate gradually decreases to a similar lower level. Therefore, the system mass difference becomes small at large recuperator effectiveness.

Fig. 15 displays the effects of the recuperator efficiency on electrical power for various mixture molecular weights. The recuperator conductance is taken as 600 kW/K. The total masses of different mixture molecular weights at the same recuperator efficiency are very close and hence are not shown here. The maximal power output decreases as the mole fraction of xenon in the mixture increases, and the maximum values occur at recuperator efficiencies between 35 % and 42 % for a 600 kW/K design. The pressure loss in the reactor core is increased as the mole fraction of xenon in the mixture grows. As a result, the maximal power shifts towards higher recuperator effectiveness where the mass flow rate is smaller to reduce the pressure drop. Also, the drop in the specific heat capacity of the He-Xe mixture reduces the power output.

For the 8 MWe design, the effects of recuperator efficiency on thermal efficiency and total mass are demonstrated in Fig. 16. As the efficiency of the recuperator gets larger, the thermal efficiency grows, and the thermal efficiency reaches 41.52 % when the efficiency of the recuperator is 0.95. On the other hand, the total mass increases slowly when the efficiency of the recuperator is less than 0.7, and then it rises drastically. Given a certain mass flow rate, improving the recuperator effectiveness results in temperature growth at the reactor inlet. The reactor power is reduced due to the constant TIT, and hence the thermal efficiency is improved. Moreover, the powers as well as masses of the turbine and compressor barely change as the output power is fixed at 8 MWe. Therefore, the increase in system mass is mainly from the

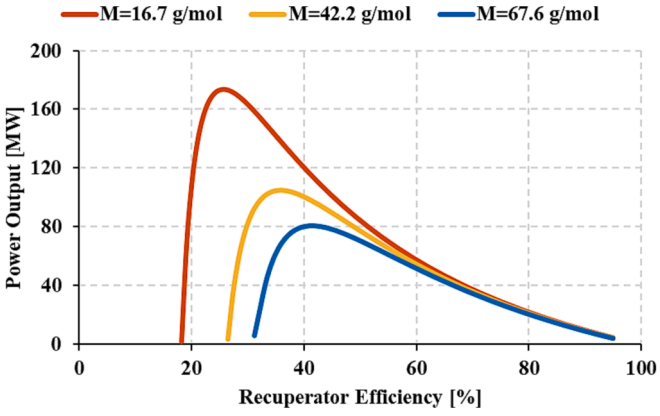


Fig. 15. Effects of the recuperator efficiency for different mixture molecular weight.

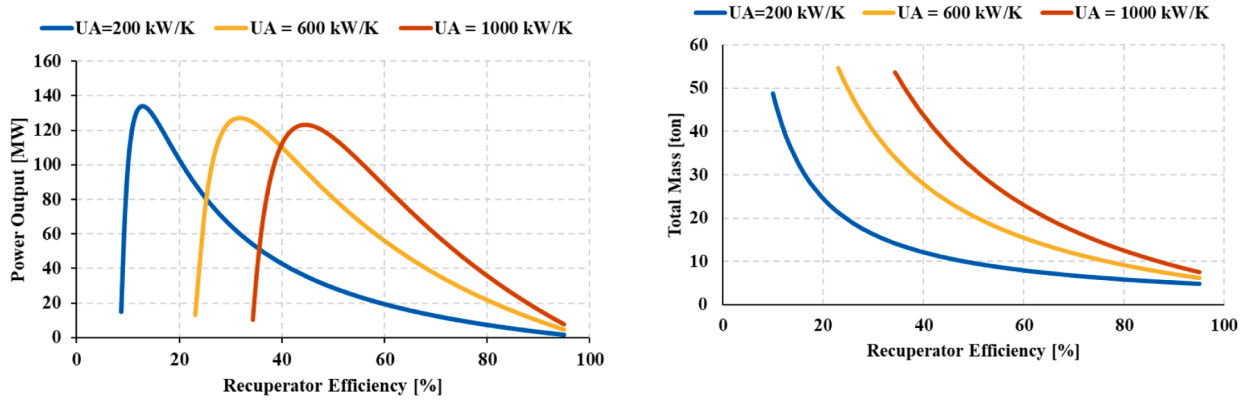


Fig. 14. Effects of the recuperator efficiency under different conductance values.

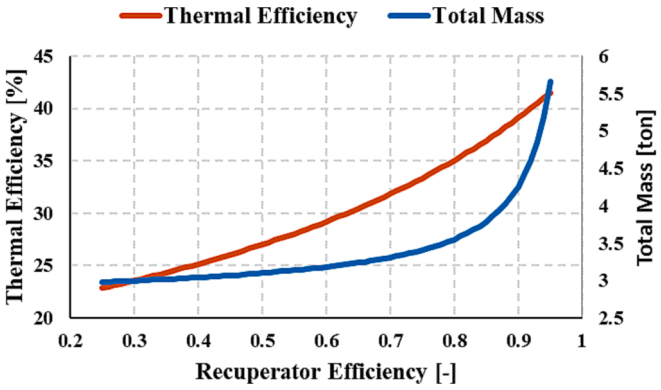


Fig. 16. Effects of recuperator efficiency for the output of 8 MWe.

recuperator. The growth in recuperator effectiveness leads to an increase in its conductance, and such an increase is nonlinear according to Equation (12)~(13).

4.3. Optimization analysis

The parametric analysis indicates that the thermal efficiency and total mass are mainly influenced by key design parameters. Among these parameters, influences of CPR, mixture's molecular weight, core height, and recuperator efficiency are non-linear and there is a trade-off between masses of different components. Therefore, a multi-objective optimization is conducted concerning these four decision variables. Their ranges of searching are displayed in Table 7. The NSGA-II algorithm [42] is used to find the Pareto frontier by minimizing the total mass and maximizing the thermal efficiency. The population size is set as 90 with a maximum generation of 300.

The results of the Pareto frontier are shown in Fig. 17. The smallest total masses at each thermal efficiency are obtained. The thermal efficiency ranges from 36.4 % to 47.7 %, and the total mass goes up with increasing thermal efficiency. At the thermal efficiency of 36.4 %, the total mass is 3.65 tons, and it increases to 6.78 tons at the efficiency of 47.7 %. Also, the growth rate of total mass increases as the thermal efficiency increases. This is due to the rapid increase in the recuperator mass.

The distribution of the decision variables in the Pareto frontier could show the optimal range of these parameters to get the smallest total mass, and they are displayed in Fig. 18. It shows that higher thermal efficiencies are accompanied by smaller CPRs, molecular weights, the height of the reactor core, and bigger recuperator effectiveness. At a lower range of thermal efficiencies, the combination of decision variables is just the opposite. The range of mixture molecular weight is 16.7 ~ 38.7 g/mol as shown in Fig. 18(b), and the corresponding mole fraction of helium is 72.5 %~90 %. As the mole fraction of helium in the mixture grows, its molecular weight is decreased, and the pressure loss in the reactor is reduced. However, the enthalpy change, as well as the mass of the turbomachinery, will be increased. Therefore, small CPRs occur at high thermal efficiency to reduce the work and mass of turbomachines as displayed in Fig. 18(a). In addition, the thermal power of the reactor core is decreased for a fixed 8 MWe design as the thermal efficiency grows. Thus, as displayed in Fig. 18(c), the required core

Table 7
Search ranges of optimization parameters.

Parameters	Compressor pressure ratio [-]	The mixture's molecular weight [g/mol]	Core height [m]	Recuperator efficiency [-]
Low limit	1.8	16.7	0.5	40 %
Up limit	3.1	67.6	1.5	95 %

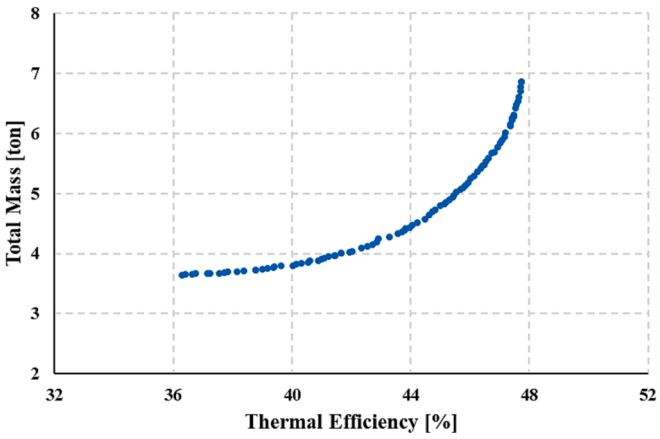
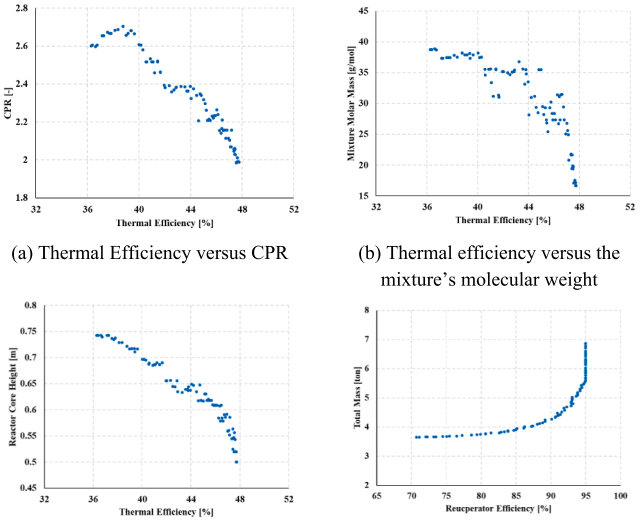


Fig. 17. The Pareto Frontiers of thermal efficiency and total mass.



(a) Thermal Efficiency versus CPR

(b) Thermal efficiency versus the mixture's molecular weight

(c) Thermal efficiency versus Core height

(d) Recuperator Efficiency versus total mass

Fig. 18. Optimal values of decision variables.

height is lowered under the constraint that the maximum temperature of the coolant tube cannot exceed 1400 K. The recuperator effectiveness ranges from 70.7 %% to 95 % as shown in Fig. 18(d), and the system mass increases dramatically when the recuperator effectiveness surpasses about 90 %, which is caused by the nonlinear growth of recuperator conductance as well as mass.

In the following analysis, point A and point B in the Pareto frontier are selected as the reference points to investigate the effects of the recuperator efficiency and the height of the reactor core on system mass near the optimal design point. The system mass difference is calculated

Table 8
Important parameters of Point A and Point B.

Parameters	Value	
	Point A	Point B
CPR [-]	2.607	2.158
M_{hexe} [g/mol]	38.22	27.32
H_{core} [m]	0.697	0.606
η_{rpc} [-]	82.61 %	95 %
$\dot{m}$ [kg/s]	94.67	77.46
Total Mass [metric ton]	3.806	5.668
η_{th} [-]	40.02 %	46.75 %

based on the value of the reference point, and Table 8 displays the parameters of the reference points.

As shown in Fig. 19, the total system mass increases when the efficiency of the recuperator for point A or the height of the reactor core for point B deviates from the optimal value. The mass of the pre-cooler almost remains the same as the reference point, which is because its conductance variation under different recuperator efficiencies is small. When the recuperator's effectiveness is increased from the optimal point, its mass grows with the increase of heat duty as shown in Fig. 19 (a). The mass of the reactor barely changes. As a result, the total mass is increased compared with the optimal point, and its growth is mainly dominated by the variation of the recuperator mass. When the recuperator's effectiveness is decreased from the optimal point, the mass of the recuperator is reduced. However, the mass of the reactor is increased due to a larger thermal power for an 8 MWe design at lower thermal efficiency. The growth in the reactor mass offsets the reduction of the recuperator mass and the total mass is increased from the optimal point. However, the magnitude of the growth in total mass is smaller compared to that when the recuperator effectiveness increases. For the decision variable of the reactor height, mass variations are displayed in Fig. 19 (b). As the height varies, it mainly affects the mass of the reactor core, and hence the variation of the total mass mainly depends on that of the reactor core. When the height of the core is reduced from the optimal point, the required core radius is enlarged. Therefore, the total mass grows significantly. What's more, further increasing the core height from the optimal value cannot achieve a smaller core radius due to the temperature limitation. As a result, the total mass increases linearly, and thermal efficiency drops slightly. The bar chart in Fig. 19 shows the mass proportion of various components in the system. For relatively lower thermal efficiencies, the pre-cooler has the largest mass share followed by the reactor core and turbo-machines. When it shifts to the higher thermal efficiencies the recuperator mass becomes the largest.

5. Conclusion

This paper presents a mass and efficiency optimization of a He-Xe gas-cooled nuclear microreactor system. The simple regenerative and intercooling cycles are constructed. Masses of the heat exchangers are calculated based on the empirical curves, and for the reactor, a minimal core radius is searched using the 1D model according to the thermal–hydraulic limitations. The effects of key design parameters on the system mass are investigated by parametric study. NSGA-II method is used to optimize the system mass in terms of the CPR, molecular weight of the He-Xe mixture, the height of the reactor core, and recuperator effectiveness. The distribution of the design variables in the Pareto Front is discussed in detail. The important specific outcomes of this study are

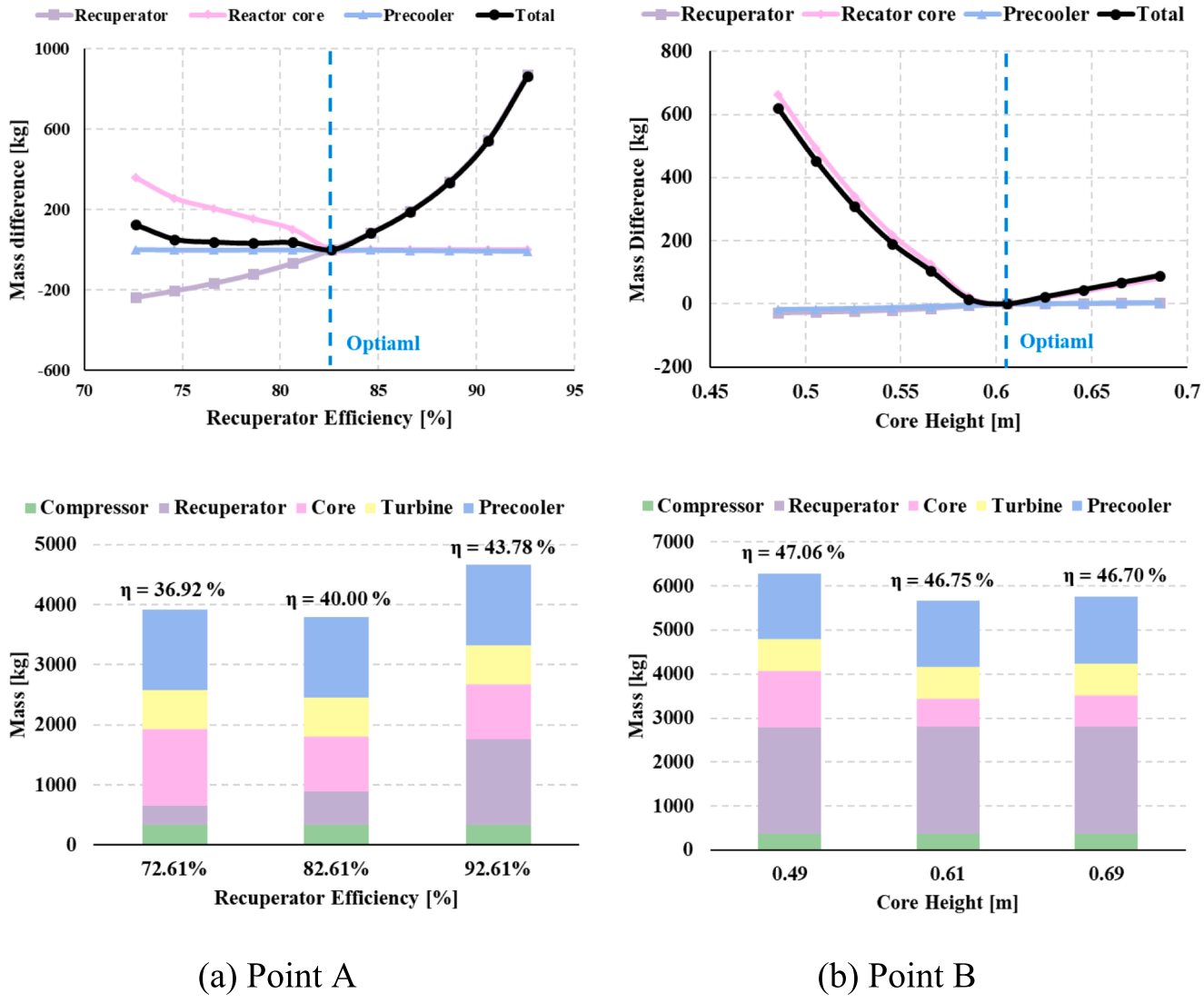


Fig. 19. Influence of recuperator effectiveness and core height on the system mass.

as follows:

(1) Increasing the mixture's molecular weight can reduce the system mass at the cost of decreasing thermal efficiency, and there is an optimal CPR for the thermal efficiency and total system mass under a set of design parameters.
(2) Increasing the height of the reactor core can reduce its mass by allowing a smaller core radius. However, larger core heights will decrease thermal efficiency due to increasing pressure loss. There is an optimal efficiency of the recuperator for detailed conductance value to maximize the electrical power, and it shifts to larger values as the mixture's molecular weight increases.
(3) The Pareto Frontier shows that at thermal efficiencies around 40 % or below, the system mass is depressed by using large CPR, the molecular weight of the mixture, the height of the reactor core, and small recuperator effectiveness. The system mass is dominated by the pre-cooler, followed by the reactor core and turbomachines. Increasing the recuperator's effectiveness from about 90 % can improve the thermal efficiency further, but its mass increases dramatically and can surpass the mass share of other components. As a result, the growth rate of system mass is significantly large when the thermal efficiency surpasses about 47 %.

CRediT authorship contribution statement

Chaoran Guan: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Xiang Chai:** Conceptualization, Methodology, Software, Writing – review & editing. **Tengfei Zhang:** Validation, Writing – review & editing. **Xiaojing Liu:** Writing – review & editing. **Hui He:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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